Name: Hafiz Waseem Ahmed
S/O: Abdul Karim
Semester: 5th
Session: 2023-2027
Dept: Computer Science
Subject: Computer Vision
Submitted to: Dr. Abdul Basit
Date: 22-6-2026

Signature: _____________

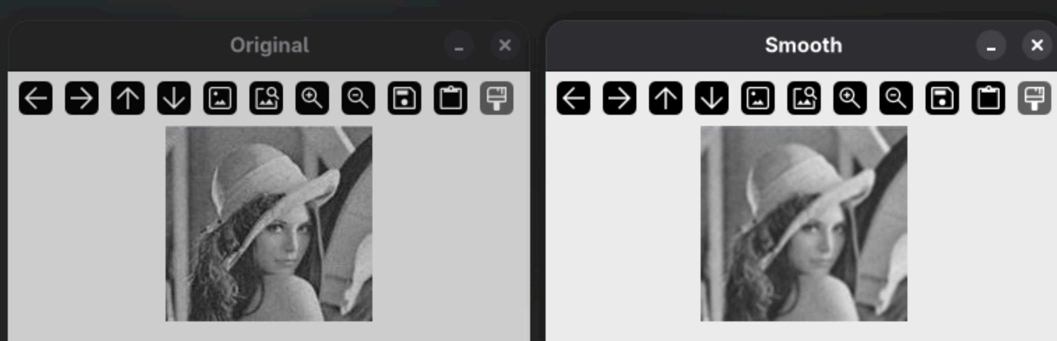
Computer Vision Assignment 4
Filtering And Edge Detection

1. Write a code to apply the following kernels.

   (a)

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \times \frac{1}{9}$$

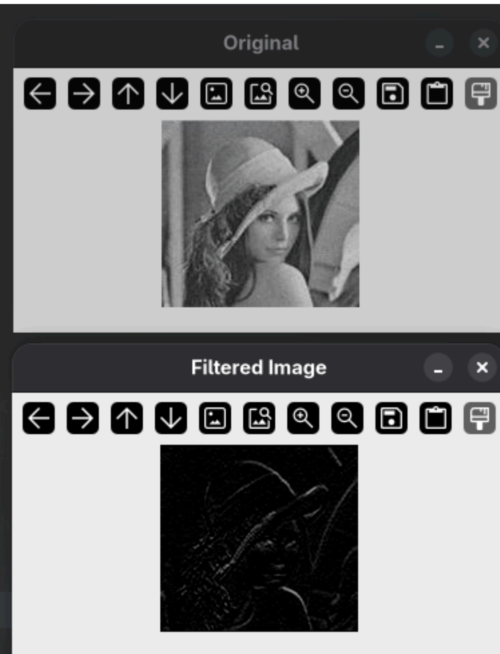
```python
import cv2
import numpy as np

kernel = np.ones((3,3)) / 1/9

img = cv2.imread("./Images/noisy.jpg")
smooth = cv2.filter2D(img, -1, kernel)

cv2.imshow("Original", img)
cv2.imshow("Smooth", smooth)

while True:
    key = cv2.waitKey(1) & 0xFF

    if key == 27:
        break

cv2.destroyAllWindows()
```



(b)

$$\begin{bmatrix} -1 & 0 & 1 \end{bmatrix}$$

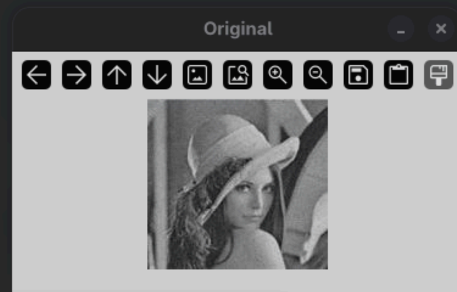
```python
import cv2
import numpy as np

kernel = np.array([-1, 0, 1])

img = cv2.imread("./Images/noisy.jpg")
filtered = cv2.filter2D(img, -1, kernel)

cv2.imshow("Original", img)
cv2.imshow("Filtered Image", filtered)

while True:
    key = cv2.waitKey(1) & 0xFF

    if key == 27:
        break

cv2.destroyAllWindows()
```

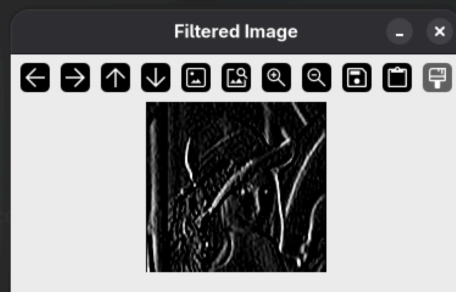


(c)

$$\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

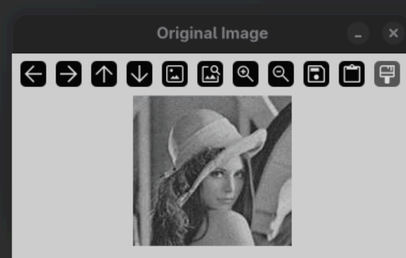
```python
kernel = np.array([
    [-1, 0, 1],
    [-1, 0, 1],
    [-1, 0, 1]])

img = cv2.imread("./Images/noisy.jpg")
filtered = cv2.filter2D(img, -1, kernel)

cv2.imshow("Original", img)
cv2.imshow("Filtered Image", filtered)

while True:
    key = cv2.waitKey(1) & 0xFF

    if key == 27:
        break

cv2.destroyAllWindows()
```

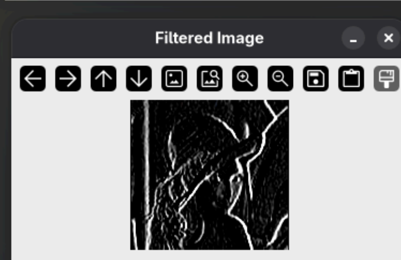




2)_Define a function that takes an image and kernel as an argument and return the resultant image.

```python
def img_filter(img, kernel):
    filtered = cv2.filter2D(img, -1, kernel)
    return filtered

print("1) fruit.png \n2) grayscale_cat.jpg \n3) japan_street.jpg \n4) lena.png \n5) noisy.jpg")
img_file_name = input("Enter the image file name: ")
img_path = "./Images/" + img_file_name

img = cv2.imread(img_path)
sobel_kernel = np.array([[1, 0, -1],
                         [2, 0, -2],
                         [1, 0, -1]])
filtered_img = img_filter(img, sobel_kernel)
cv2.imshow("Original Image", img)
cv2.imshow("Filtered Image", filtered_img)

while True:
    key = cv2.waitKey(1) & 0xFF
    if key == 27:
        break

cv2.destroyAllWindows()
```





3) Do the following.
(a) Derivate in x and y.
(b) Find the Magnitude M.
(c) Threshold the image.